CMTA
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Worldwide Plaza | 825 Eighth Avenue, FL18
New York, NY 10019 | 212.944.7722

CMTA Project No.:	B069-02-039	**Control No.:**	**H-02.00**
CMTA Project Name:	Boston Properties @ 7 Times Square - Snowflake Floors 29-31	**Page:** 1 **of** 1	

☐ **NO EXCEPTIONS TAKEN** No further review of submittal required.	☐ **REVISE AND RESUBMIT** Revise as noted and resubmit for further review.
☒ **MAKE CORRECTIONS AS NOTED** Incorporate corrections in work; resubmission is not required. Where corrections cannot be complied with, revise the exceptions and resubmit.	☐ **EXAMINED** Submittal received, formal review by others and/or formal review by this office not required. Submittal has been examined for information only.

Corrections or comments made on the shop drawings during this review do not relieve contractor from compliance with requirements of the drawings and specifications. This check is only for review of general conformance with the design concept of the project and general compliance with the information given in the contract documents. The contractor is responsible for: confirming and correlating all quantities and dimensions; selecting fabrication processes and techniques of construction; coordinating his work with that of all other trades; and performing his work in a safe and satisfactory manner.

Date:	6/15/2026	**Engineer**:	Mike Sisti
SUBMITTAL DESCRIPTION:	Pump Cuts		

COMMENTS:

CMTA Comments:

1. Provide vibration isolation for pumps and piping within 50 feet or 50 pipe diameters, whichever is greater. See specifications for details.

2. Provide valve and trim as per the details on the design documents.

3. Pumps shall meet all pressure and flow characteristics as per the design documents.

4. Contractor to provide the differential pressure at the tie in to the risers in full flow condition, as per the design documents, and provide to the manufacturer to ensure the proper head is obtained to ensure full flow throughout the system.

GRUNDFOS

Company name:
Created by:
Phone:

Date: 03/04/2026

Qty.	Description
1	**CRE 3-2 A-FGJ-A-E-HQQE**



Note! Product picture may differ from actual product

Product No.: On request

Vertical, multistage centrifugal pump with inlet and outlet ports on same the level (inline). The pump head and base are in cast iron – all other wetted parts are in stainless steel. A cartridge shaft seal ensures high reliability, safe handling, and easy access and service. Power transmission is via a rigid split coupling. Pipe connection is via combined DIN-ANSI-JIS flanges.

The pump is fitted with a 1-phase, fan-cooled, permanent-magnet, synchronous motor. The motor efficiency is classified as IE5 in accordance with IEC 60034-30-2.

The motor includes a frequency converter and PI controller in the motor terminal box. This enables continuously variable control of the motor speed, which again enables adaptation of the performance to a given requirement. The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator.

The display gives an intuitive and user-friendly interface to all functions.
The push-buttons are used to navigate through the menu structure to access pump and performance data on site and enable setting of required setpoint as well as setting of pump to "Min." or "Max." operation or to "Stop".

The Grundfos Eye indicator on the operating panel provides visual indication of pump status:

- "Power on": Motor is running (rotating green indicator lights) or not running (permanently green indicator lights)
- "Warning": Motor is still running (rotating yellow indicator lights) or has stopped (permanently yellow indicator lights)
- "Alarm": Motor has stopped (flashing red indicator lights).

Communication with the pump is also possible by means of Grundfos GO Remote (accessory). The remote control enables further settings as well as reading out of a number of parameters such as "Actual value", "Speed", "Power input" and total "Power consumption".

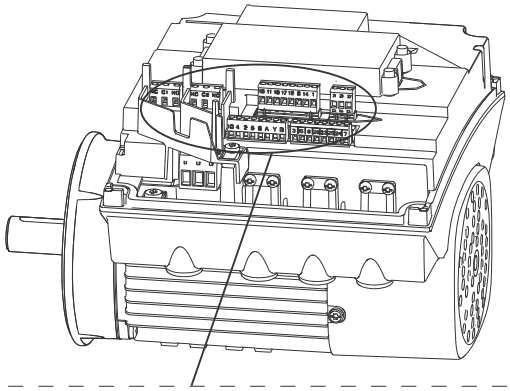
The terminal box has a number of inputs and outputs enabling the motor to be used in advanced applications where many inputs and outputs are required:

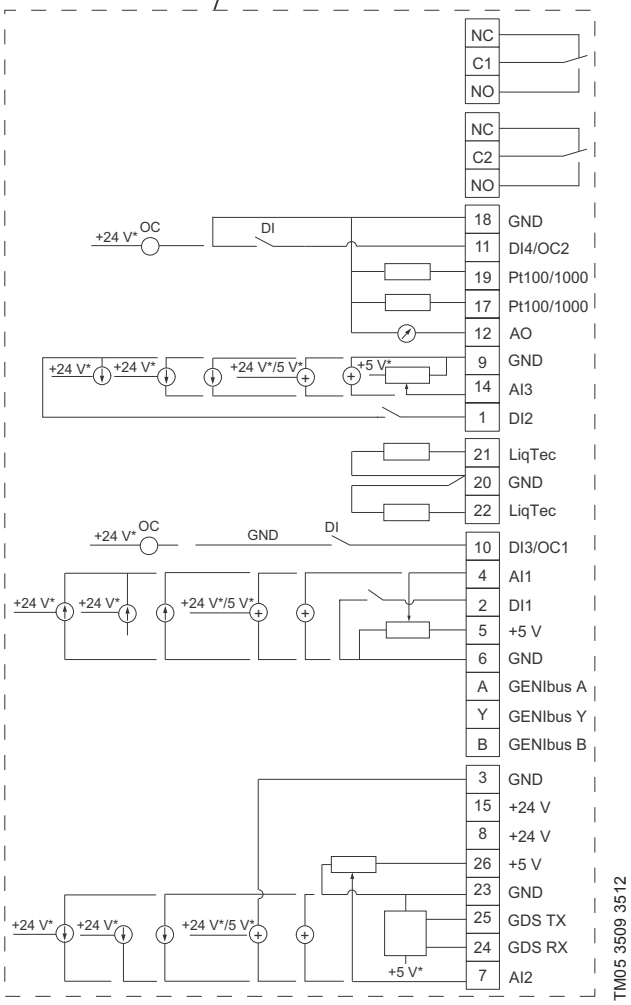
- two dedicated digital inputs
- three analog inputs, 0(4)-20 mA, 0-5 V, 0-10 V, 0.5-3.5 V
- 5 V voltage supply to potentiometer and sensor
- one analog output, 0-10 V, 0(4)-20 mA
- two configurable digital inputs or open-collector outputs
- two Pt100/Pt1000 inputs
- LiqTec, dry-running protection sensor input
- Grundfos Digital Sensor input and output
- 24 V voltage supply for sensors
- two signal-relay outputs (potential-free contacts)
- GENIbus connection
- interface for Grundfos CIM fieldbus module.

Further product details

An external sensor can be connected if controlled pump operation based on for example flow, differential pressure or temperature is required.

The operating panel on the motor terminal box features a four-inch TFT display, push-buttons and the Grundfos Eye indicator.





* If you use an external supply source, there must be a connection to GND.

Fig. 8 Connection terminals, CRE, CRIE, CRNE, CRKE, SPKE, MTRE and CME pumps

Terminal	Type	Function
NC	Normally closed contact	Signal relay 1 (LIVE or PELV)
C1	Common	
NO	Normally open contact	
NC	Normally closed contact	Signal relay 2 (PELV only)
C2	Common	
NO	Normally open contact	
18	GND	Ground
11	DI4/OC2	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
19	Pt100/1000 input 2	Pt100/1000 sensor input
17	Pt100/1000 input 1	Pt100/1000 sensor input
12	AO	Analog output: 0-20 mA / 4-20 mA 0-10 V
9	GND	Ground
14	AI3	Analog input: 0-20 mA / 4-20 mA 0-10 V
1	DI2	Digital input, configurable
21	LiqTec sensor input 1	LiqTec sensor input (white conductor)
20	GND	Ground (brown and black conductors)
22	LiqTec sensor input 2	LiqTec sensor input (blue conductor)
10	DI3/OC1	Digital input/output, configurable. Open collector: Max. 24 V resistive or inductive.
4	AI1	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V
2	DI1	Digital input, configurable
5	+5 V	Supply to potentiometer and sensor
6	GND	Ground
A	GENIbus, A	GENIbus, A (+)
Y	GENIbus, Y	GENIbus, GND
B	GENIbus, B	GENIbus, B (-)
3	GND	Ground
15	+24 V	Supply
8	+24 V	Supply
26	+5 V	Supply to potentiometer and sensor
23	GND	Ground
25	GDS TX	Grundfos Digital Sensor output
24	GDS RX	Grundfos Digital Sensor input
7	AI2	Analog input: 0-20 mA / 4-20 mA 0.5 - 3.5 V / 0-5 V / 0-10 V

TYPICAL WIRING DIAGRAM AND CONFIGURATION SETUP FOR CONTROL BY BMS
-START/STOP - DI1 - SET TO "EXTERNAL STOP"
-SPEED - AI1 - SET TO "SETPOINT INFLUENCE" (WHEN USING THE DONGLE) / "EXT. SETPOINT INFLUENCE" (WHEN USING THE PUMP INTERFACE)
-FEEDBACK - AO - SET TO "ACTUAL SPEED"
-RUN STATUS - OC1 - SET TO "PUMP RUNNING"
-ALARM - SIGNAL RELAY 1 - SET TO "ALARM"